



Department of Industrial Engineering
School of Engineering,
University of Management and Technology

Course Outline

Course code: **EF431**

Course title: **Industrial Facilities Design**

Program	BSIE
Credit Hours	2
Duration	One semester (F2024)
Prerequisites	NIL
Resource Person	Syed Rehan Ashraf
Counseling Timing (Room # 303, SEN Building)	Monday-Wednesday 11:00 AM to 1:30 PM Thursday 2:00 PM to 4:30 PM
Contacts	rehan.ashraf@umt.edu.pk Ext 3687

Chairman/Director signature.....

Dean's signature.....

Date.....

Course Learning Outcomes:

Location and Site selection, Facility design stages, processes, material handling equipment and analysis, Area allocation and space requirements, Flow analysis, fabrication of individual parts, total plant flow, Plant layout, Utilities Layout, Computerized facility layout and location, layout algorithms like CRAFT, ALDEP, CORELAP etc, Strategies for storages.

At the end of the course students should be able to:

1. **Priorities** Facility location, Facility organization processes, Product processes, and production sequence. C5
2. **Analyze** material and information flow for the different types of Products. C4
3. **Compare** several types of material handling/storage equipment system typical suitable applications. C4
4. **Design** Industrial Facility layout on the bases of product and production requirement on the bases of quantitative facility planning models C6

Course Learning Outcomes (CLOs) Mapping with Program Learning Outcomes (PLOs):

Semester	Course Code	Title	Course Learning Outcomes	PLO 1 Engg. Knowledge	PLO 2 Problem Analysis	PLO 3 Solution Design	PLO 4 Investigation	PLO 5 Mod. Tool Usage	PLO 6 Engr. & Society	PLO 7 Env. & Sust.	PLO 8 Ethics	PLO 9 Team Work	PLO 10 Communication	PLO 11 Proj. Mgmt.	PLO 12 Lifelong Learning
Fall-2023 (7 th)	EF 431	Industrial Facilities Design	1. Priorities Facility location, Facility organization processes, Product processes, and production sequence. C5	✓											
			2. Analyze material and information flow for the different types of Products. C4				✓								
			3. Compare several types of material handling/storage equipment system typical suitable applications. C4			✓									
			4. Design Industrial Facility layout on the bases of product and production requirement on the bases of quantitative facility planning models C6							✓					

Learning Methodology:

Classroom lectures, Case Studies, interactive, participative

Grade Evaluation Criteria

Components	Marks
Assignments /CEP (2, 5 Each,)	10
Quizzes (4, Average)	20
Mid Term Exam	25
Final Exam	50
Total	100

Recommended Books:

Textbook:

Facilities Planning, 4th Edition, 2010 by James A. Tompkins, John A. White, Yavuz A. Bozer, and J.M.A. Tanchoco, Wiley Publishers.

Reference Books:

1. Manufacturing Facilities: Location, Planning, and Design, 3rd Edition, 2008, by Dileep R. Sule, CRC Press.
2. Facility Layout and Location – An Analytical Approach. Francis, McGinnis & White. 2nd edition, Prentice Hall. ISBN: 978-0132992312

Calendar of Course Contents

Course code: EF 431

Course title: Industrial Facilities Design

Week	Course Contents	Textbook Chapter(s)	CLOs
1~2	Introduction to facility planning and design	CH # 1	1
3~4	Product, Processes and Schedule Design	CH # 2	1
5~6	Material and Information Flow	CH # 3	2
7	Support Staff and others requirement	CH # 4	2
8	Mid Term		
9~10	Material Handling Systems	CH # 5	3
11~12	Layout Planning Models and Design Algorithm	Ch # 6	3
13~14	Manufacturing System	CH #8 & 9	4
15	Quantitative Facilities planning models	CH #10	4
Final Term			

Note: Tentative Weeks,

Facilities Planning, 4th Edition, 2010 by James A. Tompkins, John A. White, Yavuz A. Bozer, and J.M.A. Tanchoco, Wiley Publishers.

Mapping of CLOs to Direct Assessments

CLOs▼	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Assignment / CEP 1	Assignment / CEP 2	Midterm Exam	Final Exam
1	✓						✓	✓
2		✓					✓	✓
3			✓		✓			✓
4				✓		✓		✓